$n=5$: $x_1=-1,\ x_5=1$, $x_3=0$, $x_2=-x,\ x_4=x$; $w_1=w_5$, $w_2=w_4$

$f(x)=1$:

$$\int_{-1}^{1} 1\,dx=2$$

$$2w_1+2w_2+w_3=2$$

$f(x)=x^2$:

$$\int_{-1}^{1} x^2\,dx=\frac{2}{3}$$

$$2w_1+2w_2x^2=\frac{2}{3}\Rightarrow w_1+w_2x^2=\frac{1}{3}$$

$f(x)=x^4$:

$$\int_{-1}^{1} x^4\,dx=\frac{2}{5}$$

$$2w_1+2w_2x^4=\frac{2}{5}\Rightarrow w_1+w_2x^4=\frac{1}{5}$$

$f(x)=x^6$:

$$\int_{-1}^{1} x^6\,dx=\frac{2}{7}$$

$$2w_1+2w_2x^6=\frac{2}{7}\Rightarrow w_1+w_2x^6=\frac{1}{7}$$

$x>0$

$$w_2x^2(1-x^2)=\frac{2}{15}$$

$$w_2x^4(1-x^2)=\frac{2}{35}$$

$$\Rightarrow x^2=\frac{3}{7}\Rightarrow x=\sqrt{\frac{3}{7}}$$

$$w_2\cdot\frac{3}{7}\left(1-\frac{3}{7}\right)=\frac{2}{15}\Rightarrow w_2\cdot\frac{3}{7}\cdot\frac{4}{7}=\frac{2}{15}\Rightarrow w_2\cdot\frac{12}{49}=\frac{2}{15}\Rightarrow w_2=\frac{49}{90}$$

$$w_1=\frac{1}{3}-\left(\frac{49}{90}\cdot\frac{3}{7}\right)=\frac{1}{10}$$

$$w_3=2-2\frac{1}{10}-2\frac{49}{90}=\frac{32}{45}$$

Узел	-1	$-\sqrt{\frac{3}{7}}$	0	$\sqrt{\frac{3}{7}}$	1
Вес	$\frac{1}{10}$	$\frac{49}{90}$	$\frac{32}{45}$	$\frac{49}{90}$	$\frac{1}{10}$